

EXAM 7 – SPRING 2015

17. (3.5 points)

Given the following financial projections for a company as of December 31, 2014:

<u>Year</u>	<u>After-Tax Net Income (\$000)</u>
2015	15,200
2016	16,100
2017	17,300
2018	18,500

Dividend payout ratio	60% of after-tax income
Company ROE	10.0%
Risk-free rate	3.2%
Market risk premium	4.0%
Company beta	0.75
Industry beta	0.89
Industry average growth rate	5.5%

a. (2.25 points)

Use the dividend discount model to value this company as of December 31, 2014.

b. (0.5 point)

Briefly describe two considerations when using an industry beta.

c. (0.75 point)

Assess the reasonableness of the company beta.

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EXAM 7 SPRING 2015 SAMPLE ANSWERS AND EXAMINERS' REPORT

QUESTION 17																					
TOTAL POINT VALUE: 3.5	LEARNING OBJECTIVE: B2: Value the equity of a P&C insurer based on its expected future dividends, its free cash flow to equity, or its expected abnormal earnings.																				
SAMPLE ANSWERS																					
Part a: 2.25 points																					
<i>Sample Answer 1</i> $k = 0.032 + 0.04 * 0.75 = 0.062$ $g = .4 * .1 = 0.04$ <table> <tr> <td>Year</td><td>Dividend</td></tr> <tr> <td>15</td><td>$15,200 * .6 = 9,120$</td></tr> <tr> <td>16</td><td>9,660</td></tr> <tr> <td>17</td><td>10,380</td></tr> <tr> <td>18</td><td>11,100</td></tr> </table> $9,120/1.062 + 9,660/1.062^2 + 10,380/1.062^3 + (11,100/1.062^4)[1 + 1.04/(.062-.04)]$ $= 447,055.92$ <i>Sample Answer 2</i> growth rate $g = \text{plowback} * \text{ROE} = (1 - 60\%) * 10\% = 4\%$ discount rate $k = 3.2\% + 0.75 * 4\% = 6.2\%$ <table> <tr> <td>Year</td><td>Dividend (\$000)</td></tr> <tr> <td>2015</td><td>$15,200 * 60\% = 9120$</td></tr> <tr> <td>2016</td><td>$16,100 * 60\% = 9660$</td></tr> <tr> <td>2017</td><td>$10,380 * 60\% = 10380$</td></tr> <tr> <td>2018</td><td>$18,500 * 60\% = 11100$</td></tr> </table> $\text{PV(Dividend)} = 9,120/1.062 + 9,660/1.062^2 + 10,380/1.062^3 + 11,100/1.062^4$ $= 34,544.88 (\$000)$ $\text{Terminal Value} = 11100 * (1 + 4\%) / (6.2\% - 4\%)$ $= 524,727.27 (\$000)$ $\text{Company Value} = (34544.88 + 524727.27/1.062^4) * 1000$ $= 447,055,915$		Year	Dividend	15	$15,200 * .6 = 9,120$	16	9,660	17	10,380	18	11,100	Year	Dividend (\$000)	2015	$15,200 * 60\% = 9120$	2016	$16,100 * 60\% = 9660$	2017	$10,380 * 60\% = 10380$	2018	$18,500 * 60\% = 11100$
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Part b: 0.5 point																					
<i>Sample Answer 1</i> May need to adjust the industry beta to reflect firm specific characteristics like: 1.) leverage / debt : equity 2.) mix of business / lines written <i>Sample Answer 2</i> 2 considerations when using industry β: mix of business: make sure to use companies with similar mix only																					

EXAM 7 SPRING 2015 SAMPLE ANSWERS AND EXAMINERS' REPORT

financial leverage: use an all-equity β reflecting business risk but not debt leverage
Part c: 0.75 point
Sample Answer 1 Since the firm's growth rate of 4% is less than the industry average of 5.5%, this suggests the firm may be less risky than the industry average, so the lower beta of .75 compared to the industry is reasonable. Sample Answer 2 Company β < Industry Company g < Industry Firm is growing slower, implying less potential for risk. It makes sense the firm's β would be smaller than the industry
EXAMINERS' REPORT
Part a
Candidates were expected to apply the dividend discount model (DDM) to estimate the value of a company. To obtain full credit, candidates were expected to clearly present the correct formulas/values for the dividends during the forecast horizon, the terminal value, and the overall firm value. Most candidates performed very well on this question. The following errors were typical of candidates who did not receive full credit: • Minor arithmetic mistakes / transcription errors • Using net income for the cash flows instead of the dividends (not applying the dividend ratio) • Using the industry beta to compute the discount rate (without sufficient justification) • Using the industry growth rate for the terminal value (without sufficient justification)
Part b
Candidates were asked to briefly describe two considerations when using an industry beta. Most candidates were able to provide at least one of these considerations, but the majority of candidates did not identify both of them. Incorrect responses commonly made reference to growth rates, investment portfolios, ROE's, and company size.
Part c
This question part required candidates to synthesize readily available information to assess the reasonableness of the company beta. Candidates generally performed very well on this part. Most candidates were able to provide an appropriate assessment. Candidates who did not receive full credit often had correct assessments with incomplete explanations. The following errors were common: • Not clearly explaining that a higher growth rate is indicative of a riskier firm (warranting a higher beta) • Providing a general explanation of the situation without clearly committing to an assessment of the reasonableness of the company beta Candidates who incorrectly calculated a higher growth rate for the company in their response to part a were not penalized for providing the opposite answer in this question part, provided that their logic was correct and sufficiently delineated.